

Nathan Espejo

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EDUCATION

Western University

Bachelor of Engineering Science in Software Engineering

London, ON

Aug. 2024 – May 2028

EXPERIENCE

VP Machine Learning

Sept 2025 – Present

3D Western

London, ON

- Serving as VP of Data/ML, leading a team of developers in model benchmarking and reinforcement learning to optimize a classifier that flags inappropriate 3D models for print approval.
- Built data generation workflows for 3D model analysis by scraping assets and creating automated rendering scripts to produce labeled image datasets for AI training.

Software Engineering Intern

May 2025 – Aug 2025

TeraGo

Markham, ON

- Built an automated Jira ticket workflow from email notifications using a local LLM (Ollama) for structured extraction, Python orchestration, and Microsoft Graph for secure message retrieval.

IT Intern

May 2025 – Aug 2025

Microcomputer Consulting Inc.

Mississauga, ON

- Configured and deployed client workstations across multiple organizations, including Windows imaging, account setup, and environment-specific software configuration.

PROJECTS

Local AI Platform | *Local LLMs, RAG, docling, Firecrawl, MCP, Open WebUI*

Apr 2026

- Deployed local LLMs (text, vision, speech) with hybrid routing to cloud providers, Firecrawl powered web search, and MCP based agents with custom skills enabling private AI automation without relying on third party APIs.
- Built a document ingestion pipeline using docling to extract text, tables, and formulas from PDFs, with embedding, and reranking into a vector store for precise retrieval-augmented generation in knowledge based workflows.

Homelab Infrastructure | *Docker, Linux, Tailscale, Caddy, UFW*

Apr 2026

- Set up self-hosted services with a centralized Docker Compose infrastructure with automatic TLS and VPN-routed downloading, enabling secure remote access from any network through a single reverse proxy endpoint.
- Hardened the infrastructure by binding admin interfaces to localhost behind the reverse proxy and enforcing a default-deny firewall, eliminating the external attack surface without sacrificing usability.

AgenticArmy | *TypeScript, Python, FastAPI, VS Code Extension*

Jan 2026

- Built a human-in-the-loop multi-agent coding workflow with a VS Code extension and backend coordinator for planning, conflict analysis, and QA.
- Implemented orchestration patterns including approval gates, state transitions, event-driven coordination, and simulation mode for reliable agent execution.

Auralis | *Next.js, React, Three.js, FastAPI, face-api.js*

Nov 2025

- Built an AI-powered virtual doctor with emotion-aware consultations and report generation, winning **Best AI Application Built with Cloudflare**.
- Developed emotion mismatch detection and adaptive response logic using facial signals and conversational context to improve safety-focused interactions.

TECHNICAL SKILLS

AI Systems & Agentic Coding: Multi-Agent Orchestration, MCP, RAG, Local/Hybrid LLM Workflows, Model Routing, Human-in-the-Loop, Cursor, Copilot, Claude Code, Firecrawl, docling, Bifrost

Languages, Web & Data: Python, TypeScript, JavaScript, C#, C++, Java, SQL, Bash, React, Next.js, FastAPI, Node.js, PyTorch, TensorFlow, scikit-learn, pandas

Infrastructure & Automation: Linux (Arch, Debian), Docker, Tailscale, Git, Caddy, UFW, Server Automation, Remote Access

Hardware, XR & Embedded: Unity3D, C#, VR, AR, Arduino, ESP32, Raspberry Pi, PlatformIO, 3D Printing, CAD, OnShape, Blender